Mechanics of update_by_query

Elasticsearch first parses the input query and identifies the shards that contain documents that could potentially match our query. For each of these shards, Elasticsearch executes the query and finds all documents that match it.

Elasticsearch will run the script we've provided on each of these matched documents. Before updating the documents, Elasticsearch checks if the current document's version corresponds with the version when it was found during the query phase. If they match, the update proceeds; if not (perhaps, the document was updated due to another operation interim), Elasticsearch retries the operation. Each document is then updated in memory and re-indexed, marking the old document as deleted while a new one added to the index.

When updating a particular document fails, the failure is logged but the rest of the documents will get updated as per the script. Any conflicts during the update process can be either retried a specific number of times or ignored completely as per the preconfigured settings. Once the operation is completed and the indexes are refreshed, so the changes are available for search.

The `_update_by_query` operation returns a response that includes count of total processed documents, version conflicts, successfully updated ones and failed documents. As you can imagine, the `_update_by_query` operation is a resource-intensive operation, hence careful consideration of its affect on your cluster's performance needs to be considered. We can let Elasticsearch process the updates in batches if it's using too many resources. The batch size and throttling limits can be configured.

We now know the mechanism to update documents. It is time to discuss delete operations—specifically, how to delete a single document or multiple documents simultaneously. The next section is dedicated to deleting documents in multiple ways.

5.5 Deleting documents

When we want to delete documents, there are two methods: using an ID or a query. In the former case, we can delete a single document; and in the latter, we can delete multiple documents at once. When deleting by query, we can set filter criteria (for example, delete documents whose status field is unpublished or documents from last month). Let's see both of these methods in action.

5.5.1 Deleting with an ID

We can delete a single document from Elasticsearch by invoking the HTTP `DELETE` method on the indexing document API:

```
DELETE <index_name>/_doc/<id>
```

The URL is the same as the one we used for indexing and retrieving the document. Given the ID of the document to be deleted, we construct the URL by specifying the